



Overview of graphite supply chain and its challenges

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Abstract

The importance of graphite is increasing with the global shift toward carbon neutrality and clean energy transitions. As a critical mineral for battery anodes, graphite demand is projected to grow fourfold by 2030 compared to 2023 levels, with overall demand doubling during the same period. However, graphite is classified as a high-risk mineral due to its heavily concentrated supply chain, which poses significant geopolitical risks. The globalization of manufacturing and trade has further exacerbated supply chain vulnerabilities by centralizing production in low-cost regions, deepening dependence on specific countries. This letter provides a comprehensive overview of the graphite supply chain and its associated challenges while offering strategic insights for stabilization and diversification.

Keywords Graphite · Supply chain · Flake graphite · Anode material · Clean energy transition

1 Introduction

The global transition to clean energy demands swift and decisive action to combat climate change. Renewable energy systems and electric vehicles (EVs) depend on substantial quantities of critical minerals such as lithium, nickel, cobalt, copper, graphite, and rare earth elements (REEs) to produce high-performance batteries, wind turbines, and other low-carbon technologies (IEA, 2024; IRENA, 2024). Lithium is crucial for battery electrolytes, nickel, and cobalt are integral to battery cathodes, graphite forms battery anode, and REEs play key roles in advanced electric motors and

semiconductors. Notably, EV requires about five times more critical minerals than internal combustion engine vehicles (IEA, 2024), meaning the increase in EV adoption directly drives soaring demand for the critical minerals. Although these minerals have historically played a minor role in global markets, they are rapidly becoming indispensable for achieving carbon neutrality, reducing greenhouse gas emissions, and enhancing energy security (Woodley et al., 2024).

Among these critical minerals, graphite stands out as the primary anode material in lithium-ion batteries due to its excellent electrical conductivity, layered carbon structure for efficient ion intercalation, and comparatively low cost (Zhang et al., 2021). As shown in Fig. 1, graphite's unique carbon layers allow lithium ions to move in and out during charging and discharging, thereby enabling reliable energy storage. With battery demand soaring for EVs and stationary energy storage, graphite is under intense supply pressure, especially since it currently has no effective substitute in battery anodes (IRENA and NUPI, 2024a).

Despite graphite's strategic importance to the clean energy transition, its supply chain faces significant risks mainly due to its highly concentrated production, primarily driven by low economic viability in countries outside China (IEA, 2024). Although China's manufacturing boom over the past five decades has fueled supply growth, graphite's geological characteristics also contribute to its intense concentration compared to other minerals. Because graphite often occurs in small and scattered deposits, large-scale

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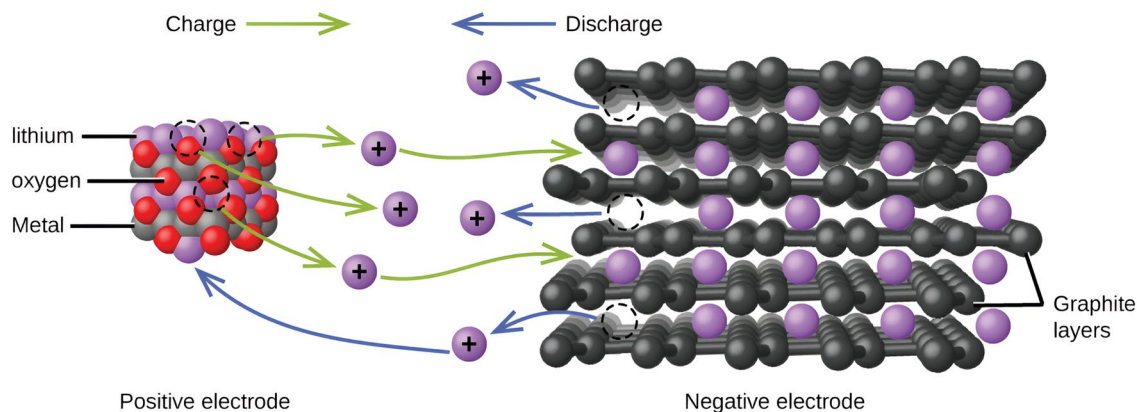


Fig. 1 Mechanism of graphite anode material in battery (Davison and Van Rythoven, 2023)

mining and associated economies of scale are difficult to achieve. In contrast, commodities like iron ore or coal are mined using advanced, large-scale equipment, allowing for more profitable operations. For context, while global coal production exceeds eight billion tons per year, graphite production remains at approximately 1.2 million tons (USGS, 2024).

Furthermore, graphite's relatively low production volume is not offset by a higher market price. The lower value of graphite compared to other critical minerals has discouraged extensive investment in graphite supply. Although the graphite anode constitutes about 30% of a battery cell's weight, it represents only around 12% of the total cost (Fastmarkets, 2024), underscoring graphite's limited economic appeal. Prior to the surge in battery demand, graphite was considered a low-profit resource due to its narrow industrial applications. In the mid-twentieth century, nations such as Austria and South Korea were major producers (Choi & Ryu, 2020); however, globalization and trade liberalization gradually shifted production to developing regions with lower costs, further concentrating the supply chain in areas with competitive economic advantages.

Over the past five decades, most countries outside China have significantly reduced their graphite production capacities. Meanwhile, China's abundant, high-quality deposits and lower production costs have cemented its dominance in the graphite supply chain. This concentration has heightened geopolitical risks, particularly as demand continues to rise with the growing need for batteries.

In this letter, we provide an overview of the graphite supply chain and the current supply–demand dynamics. We then offer strategies and insights from geological and market perspectives to diversify and reinforce graphite supply, ensuring enhanced resilience and long-term sustainability for the clean energy era.

2 Characteristics of graphite

Carbon exists in various forms across the Earth's crust, atmosphere, oceans, and organic matter. In the Earth's crust, 80–90% of carbon is found as carbonate minerals, while graphite accounts for only about 0.5% of crustal carbon. Due to its remarkable stability and chemical inertness, graphite remains isolated from the carbon cycle (Robinson et al., 2017).

Most natural graphite forms through high-temperature thermal metamorphism of organic matter found in sedimentary rocks or subsurface organic deposits. In modern times, synthetic graphite is also widely produced. The English term “graphite” originates from the Greek word *graphein*, meaning “to write,” highlighting its historical use in writing materials.

The quality of graphite is typically expressed as TGC (Total Graphitic Carbon) or Cg (Graphitic Carbon), which represent the percentage of graphite in a given sample. For production and trade, the commonly used measurement is GC (graphitic carbon) mass, which is based on the actual amount of graphite contained in the product.

Graphite can be categorized into two main types: natural graphite and synthetic graphite. Battery-grade graphite includes natural graphite and synthetic graphite, each with unique characteristics. Major characteristics of graphite are listed in Table 1. Natural graphite is less expensive but has relatively lower charge–discharge efficiency and a theoretical capacity of 372 mAh/g. Synthetic graphite, on the other hand, is more purified, better suited for fast charging/discharging, but has a slightly lower theoretical capacity of 362 mAh/g (Zhao et al., 2022). Due to these complementary properties, secondary batteries for vehicles often use a mix of both.

Table 1 Characteristics of graphite by classifications (Fastmarkets, 2024; Robinson et al., 2017; USG, 2024)

Graphite type	Natural graphite		Synthetic	
	Amorphous	Flake	Vein	
Crystallinity	Microcrystalline	Crystalline	Crystalline	Microcrystalline to crystalline
Origin	Contact metamorphism, often by diagenetic or granitic intrusions, and (or) regional metamorphism of carbonaceous sediments, often coal	Regional metamorphism of carbonaceous sediments at or exceeding amphibolite facies conditions	Epigenetic veins and lodes formed from metamorphic fluids in high-grade metamorphic rocks, usually granulites	Produced by heat treatment (graphitization) of, or chemical deposition from, hydrocarbon materials above 3000 °C
Orebody	Layers, seams, and lenses in carbonaceous rock; may be folded and faulted	Strata-bound; tabular or lens form. Irregular in hinge areas of folds	Vein and fracture-filling within or cross-cutting metamorphic structures and rock contacts	NA
Ore grade (% TGC)	25–60 (locally higher than 90)	5–30 (locally higher than 60)	40–90	NA
Product (concentrate) grade (% TGC)	60–90	75–95 (95% for battery)	90–99.9	99.95
Major producers	China, Mexico	China, Madagascar, Mozambique, Canada, Australia	Sri Lanka	China, Japan, South Korea
Main uses	Refractories, steel industry, paint	Battery, Refractories, brake linings, lubricants, expandable graphite	Carbon brushes, brake linings, and lubricants	Batteries, carbon brushes, graphite electrodes, nuclear moderator rods
Avg. price (\$/tonne)	565	1200 (low-end) 4000–6000 (Battery AAM)	2500	2000–4000 (low-end) 6000–10,000 (Batter AAM)

2.1 Natural graphite and its formations

Natural graphite is an allotrope of carbon, characterized by its crystalline structure with carbon atoms arranged in hexagonal layers. It forms under specific geological conditions, typically in regions subjected to high-temperature and high-pressure processes such as regional and contact metamorphism. Natural graphite occurs as three primary types: amorphous, flake, and vein type. For lithium-ion battery anode, only flake type graphite is used.

2.1.1 Amorphous graphite

Amorphous (also called microcrystalline or cryptocrystalline) graphite deposits are generated through the metamorphism of coal, petroleum reservoirs, or other high-carbon sedimentary rocks. These deposits consist of aggregates of fine-grained graphite crystals. Amorphous graphite forms at temperatures above 300–400 °C (Landis, 1971), and the quality (grade) of the graphite depends on the characteristics of the precursor carbonaceous material. Deposits generally contain 25–60% carbon, but highly metamorphosed coal seams can yield up to 85% graphite carbon (Taylor, 2006). These deposits appear as layers, seams, or lenses, often several meters thick and extending up to one kilometer in length (Robinson et al., 2017).

Major global deposits of amorphous graphite are located in Sonora, Mexico; Siberia, Russia; Austria and Ukraine in Europe; and the coal-rich regions stretching from China to the Korean Peninsula (Taylor, 2006).

2.1.2 Flake graphite

Flake (or crystalline) graphite deposits are predominantly found in metamorphic rocks such as schist, gneiss, quartzite, and marble. These deposits form when carbonaceous material in host rocks transforms into graphite during regional metamorphism. Typically, this occurs under amphibolite facies metamorphic conditions, leading to the crystallization of well-aligned order and recrystallization of the rock matrix (Landis, 1971).

Flake graphite deposits are typically found in the roots of continental mountain belts, where post-metamorphic uplift and erosion are necessary to bring these deposits to mineable level (Robinson et al., 2017). These deposits often appear as strata-bound lenses or layers, with individual lenses reaching several tens of meters in thickness and hundreds of meters in length. The graphite flakes in these rocks are oriented parallel to the plane of foliation (Taylor, 2006). Carbon content varies widely, from 8 to

15% in most deposits to as high as 60% in certain areas, such as Madagascar (Robinson et al., 2017).

Graphite flakes, categorized by size, serve various applications. 150–300 + μm (“Jumbo/Large Flake”) and 75–150 μm (“Fine Flake”) are common, with the only fine flake favored in battery anode (Fastmarkets, 2024). Larger flakes are premium due to their use in refractories, where size and purity enhance oxidation resistance. Larger, purer flakes improve durability and extend refractory life (Imerys, 2016).

Notable flake graphite deposits are found in Brazil, Canada, China, and East Asia. In particular, the Jixi district in Heilongjiang Province, China, and Madagascar are known for their world-class, high-quality flake graphite resources (USGS, 2024).

2.1.3 Vein graphite

Vein (or lump) graphite has a crystalline structure and forms as veins in high-temperature regional metamorphic environments. These deposits are associated with igneous intrusions into metasedimentary belts, where graphite veins develop along metamorphic rock structures, rock contacts, or folds. Vein graphite deposits are typically high-grade, containing 40–90% or more graphite (Robinson et al., 2017).

Sri Lanka hosts the most economically significant vein graphite deposits. In these deposits, carbon is transported by metamorphic fluids through oxidation/reduction reactions, cooling, or mixing processes, leading to the formation of high-purity graphite veins (Touret et al., 2019).

2.2 Synthetic graphite

Synthetic graphite is manufactured from petroleum by-products like needle coke and pitch through a high-temperature graphitization process at approximately 3000 °C. This process produces highly purified and refined graphite, making synthetic graphite more widely used than natural graphite in many applications (Lee et al., 2015). However, the graphitization process consumes substantial energy, resulting in significant carbon emissions. This characteristic poses a challenge in meeting increasingly stringent ESG (Environmental, Social, and Governance) standards, which emphasize sustainability and low carbon footprints (Vohler et al., 2021).

3 Supply and demand of graphite

3.1 Demand of graphite

With the rapid growth in battery demand, the need for graphite is projected to increase substantially due to the current

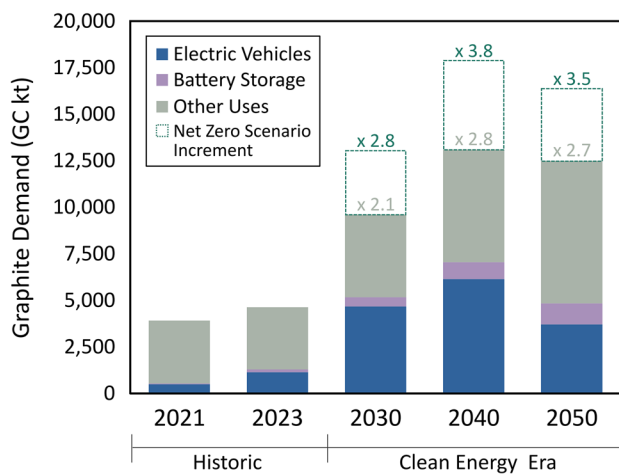


Fig. 2 Change in demand of graphite (including natural and synthetic) in clean energy era, based on conservative scenario and net-zero scenario proposed by IEA (2024)

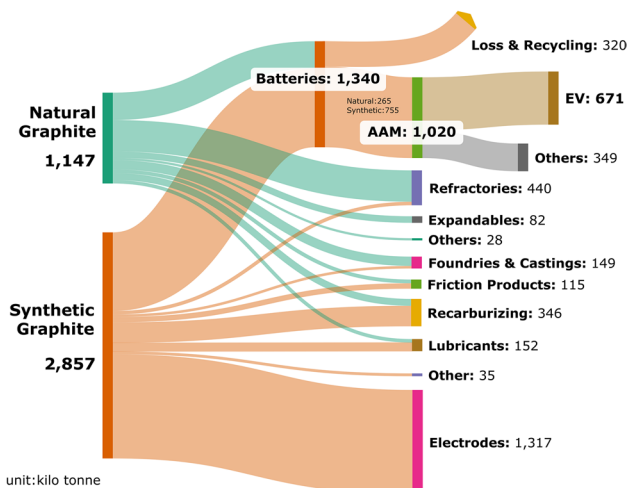


Fig. 3 Demand of graphite by applications (data from Fastmarkets, 2024)

absence of widely available alternatives (Fig. 2). By 2030, the demand for battery-grade graphite is anticipated to quadruple compared to 2023 levels, driving an almost twofold rise in total graphite demand over the same period (IEA, 2024).

Graphite is used in a wide range of applications, which makes its supply chain increasingly complex as it moves closer to the final product. As shown in Fig. 3, graphite has traditionally been used in refractories, carburizers, and electrodes for electric arc furnaces. However, its use in battery anodes has grown significantly in recent years. Batteries now account for approximately one-third of total graphite demand, with about half of this used in electric vehicle (EV) batteries. Each EV battery requires 50–100 kg of graphite

(Olson et al., 2016), and the rapid adoption of EVs is driving a sharp increase in graphite demand.

3.2 World reserve of graphite

Graphite is found and produced in many countries around the world. As shown in Fig. 4, China has the largest reserves of graphite, including both flake graphite and amorphous graphite, followed by Brazil, Mozambique, and Madagascar. In China, 85% of the graphite reserves are flake graphite, while 15% are amorphous graphite. In China, two third of flake graphite is located in Heilongjiang (47%) and Inner Mongolia (19%), respectively (Cui et al., 2017).

Among African countries, Madagascar, Mozambique, and Tanzania in East Africa hold significant graphite reserves, most of which are flake graphite. Production in these regions is expected to grow rapidly in the coming years due to increasing global demand.

3.3 World production of flake graphite

The primary producers of flake graphite, which is used in batteries, are China, Mozambique, and Madagascar. Table 2 lists the graphite production volume in tonnes of Total Carbon (TC) for each country in 2023, along with the names of operating mines and major upcoming projects (Fastmarkets, 2024).

China produced 800,000 tonnes of natural graphite ore, accounting for approximately 77% of global production, with the top three countries contributing a combined 97% of the total (Fastmarkets, 2024). Additionally, new mining projects are underway in countries like Tanzania, Canada, and Australia, which are expected to expand graphite production in the near future.

3.4 Graphite deposits and production in South Korea

South Korea was a significant producer of graphite during the 1950s and 1960s; however, its production has declined markedly in recent decades (Choi & Ryu, 2020). As illustrated in Fig. 5, graphite was historically mined across the country, with the exception of the Gyeongsang Basin region in the southeast (KORES, 2020).

Flake graphite hosted within metamorphic sedimentary rocks, such as gneiss or crystalline schist, is considered economically viable in South Korea (Lee et al., 2007). These flake graphite deposits typically exhibit grades of 4–20% total graphitic carbon (TGC), which can be upgraded to approximately 70% TGC through beneficiation processes. However, most deposits are characterized by low graphitization levels and are classified as intermediate materials, between anthracite and graphite (KORES, 2020).

Fig. 4 World graphite reserve distribution by country (data from USGS, 2024)

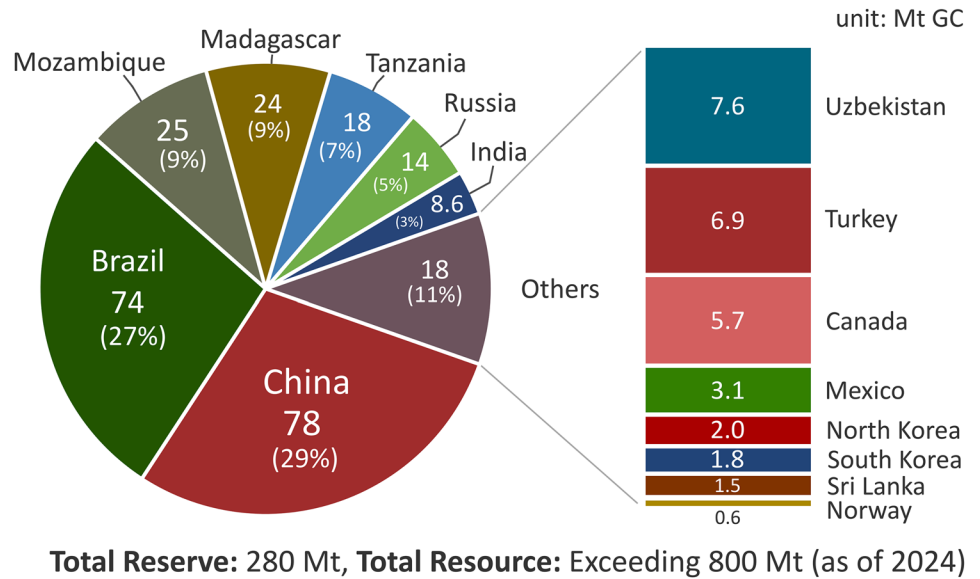


Table 2 Production of graphite by country (Fastmarkets, 2024)

Country	Currently operating mine	Production in 2023 (TC tonne)	Upcoming projects
China	Various in Heilongjiang and Inner Mongolia Province	800,000	Various
Mozambique	Ancuabe Balama	100,000	Nipepe
Madagascar	Antsiracambo Marovintsy Molo Sahamamy Vatomina	78,500	Maniry
Brazil	Maiquinique Pedro Azul Salto da Divisa	24,000	Santa Cruz Okanjande
Russia	Taiginsky	10,000	
Tanzania	Tanxin	8000	Mahenge Nachu Bunyu Epanko Lindi Jumbo
Norway	Traelen	6500	
Canada	Lac des Iles	5000	Matawinie Uatnan Bissett Creek
Vietnam	Yen Bai	4000	
Ukraine	Zavalievsky	1600	
Others		—	Vittangi (Sweden) Siviour (Australia) Munglinup (Australia) Kasiya (Malawi) Graphite Creek (US)
Total		1,037,600	

Rapid urbanization and the availability of cheaper imported raw materials have significantly diminished the profitability of domestic mining operations. As of 2022,

reports indicated minimal production from one flake graphite mine and one amorphous graphite mine, though the operational status of these sites remains unclear. By 2023,

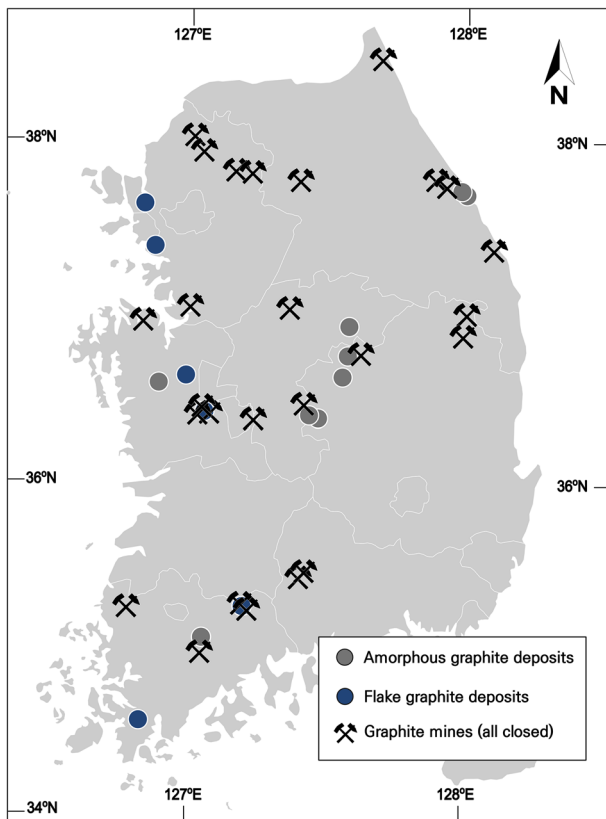


Fig. 5 Distribution of amorphous and flake graphite deposits and former mining sites (data from KORES, 2020)

no active graphite production was reported in South Korea (KIGAM, 2024).

4 Supply chain of graphite

Graphite is used in various applications, leading to different supply chain configurations depending on the end product. This section focuses on the supply chain for graphite used in battery anode materials, where supply risks are most prominent.

The supply chain for natural graphite can be divided into six stages: mining, beneficiation, purification, spheroidization, coating, and active anode material (AAM) manufacturing (Fig. 6). Among these, mining and beneficiation are considered upstream processes in the battery supply chain, while purification, spheroidization, and coating are classified as midstream processes. The final product, coated spherical graphite, is used directly as AAM in batteries.

Synthetic graphite is manufactured by replicating the high heat and pressure conditions under which natural graphite forms. It is produced using by-products of petroleum such as needle coke and pitch. This process involves intense energy consumption, particularly during graphitization, to create a material with high purity and uniformity suitable for battery anode.

4.1 Supply chain of natural graphite

4.1.1 Mineral exploration

The exploration of natural graphite primarily leverages its distinct electrical properties, particularly its relatively high electrical conductivity compared to other minerals. The techniques such as electromagnetic (EM), self-potential (SP), and induced polarization (IP) surveys are widely recognized for their effectiveness in identifying graphite deposits (Dentith & Barrett, 2003; Riva et al., 2019). In contrast, gravity surveys are less commonly utilized due to graphite's density being comparable to that of surrounding host rocks, making differentiation challenging unless associated with sulfide minerals (Scott, 2014). Furthermore, graphite deposits rarely occur in massive, homogenous forms; instead, they are often vertically distributed within complexly folded geological structures, further reducing the utility of gravity-based methods. Similarly, magnetic susceptibility surveys are infrequently applied, as graphite's magnetic properties do not remarkably differ from those of the host rocks.

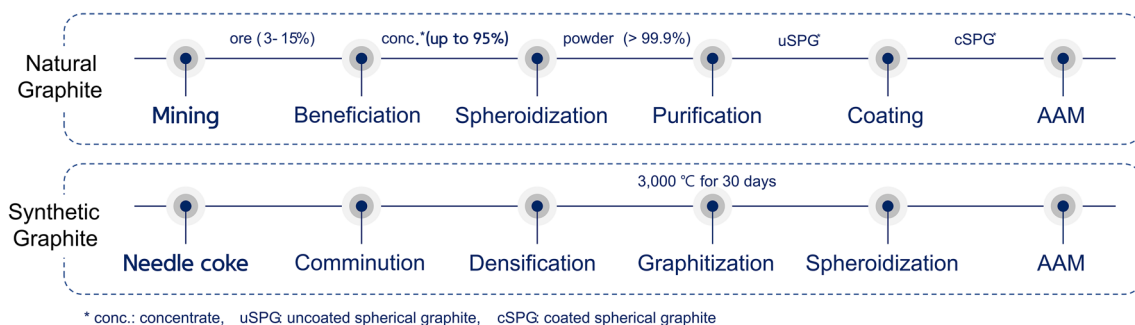


Fig. 6 Supply chain of natural graphite and synthetic graphite

4.1.2 Mining

Graphite deposits are commonly found as relatively small lenses or inclined sheets, which naturally constrain the scale of open-pit mining operations. Graphite extraction is typically performed using either surface or underground mining techniques, with surface mining being more prevalent. The grade of graphite ore varies significantly depending on the mining method employed. Surface mines usually produce graphite with grades ranging from 3 to 15% total graphitic carbon (TGC), whereas underground mines often yield higher-grade deposits, with TGC content ranging from 10 to 90%.

Surface mining is favored for its cost-effectiveness and high productivity. The primary surface mining methods include open-pit and stripping mining. Prominent examples of open-pit mines are the Balama mine in Mozambique, the Santa Cruz mine in Brazil, and the Molo mine in Madagascar. In contrast, underground mining, while more expensive and associated with increased safety risks for workers, offers distinct advantages for the extraction of high-grade deposits and has a reduced environmental footprint. Effective underground mining methods for graphite include sublevel (or long-hole) stoping and cut-and-fill mining. Notable examples of underground graphite mines include the Traelen mine in Norway and the Ragedara mine in Sri Lanka.

4.1.3 Beneficiation

Graphite refining is typically conducted using conventional beneficiation techniques, including crushing, grinding, gravity separation, and flotation. These processes effectively enhance the purity of graphite to levels between 80 and 95% total graphitic carbon (TGC). To meet the stringent requirements for battery-grade graphite, further refinement during this stage increases the purity to 95% TGC.

Gravity separation exploits the low density of graphite particles, employing equipment such as jigs to achieve material separation. Flotation processes utilize the hydrophobic nature of graphite surfaces to isolate it from other minerals (Gao et al., 2024). For larger flake graphite particles, reverse flotation is often preferred due to its efficiency in preserving flake size and integrity. Conversely, lower-grade fine graphite is typically refined using methods such as film flotation, which are more effective for smaller particles (Chehreh Chelgani et al., 2016).

4.1.4 Further processing for battery anode

The production of battery anode materials necessitates the enhancement of both surface properties and purity of graphite through specialized processing techniques such as spheroidization, purification, and coating. The majority of

production facilities for these processes are concentrated in China. Major producers of natural graphite anode materials include BTR, XFH, and Shanshan in China, as well as Vidalia in the United States and POSCO in South Korea.

Spheroidization is a key step in this process, wherein flat flake graphite particles are reshaped into spherical forms with diameters typically ranging from 15 to 20 μm . This transformation improves charge–discharge performance by enhancing alignment within the graphite layers and increasing packing density (tab density) during anode assembly (Fischer et al., 2023). Spheroidization is an essential step to manufacture AAM, however, the yield after the process only reaches 30% in average (Jara et al., 2019).

Following spheroidization, the graphite, with an initial purity of 95% graphitic carbon (gC), undergoes additional refinement to achieve a purity level of 99.9% or higher. This critical step removes silicate and metallic impurities embedded within the graphite structure. The removal of silicates often employs hydrofluoric acid, a highly effective but environmentally hazardous reagent, posing significant challenges in sustainable manufacturing.

To further improve performance, a pitch coating is applied to the graphite's surface. This coating extends battery life by protecting the graphite's layered structure, reducing the risk of structural degradation during repeated charge–discharge cycles (Jara et al., 2019).

4.2 Supply chain of synthetic graphite

Synthetic graphite is made by mixing coke, a by-product of petroleum or coal, with 15–20% binder pitch. This mixture is shaped and then heat-treated through a graphitization process at approximately 3000 $^{\circ}\text{C}$. The result is a layered crystal structure of carbon.

The graphitization process is the most time-consuming and expensive step in synthetic graphite anode production, taking 15–30 days to complete. This includes heating (around 2 weeks), graphitization (2 days to 1 week), and cooling (2 weeks). These stages require substantial energy, making it a major cost driver in production.

The leading synthetic graphite anode manufacturers are BTR, Shanshan, and Zichen from China, together accounting for 40% of global shipments (SNE Research, 2024).

4.3 Recycling

Graphite recycling involves pretreatment, recovery, and reuse. Pretreatment includes battery shredding, which separates components like anodes, cathodes, and collectors via methods such as flotation and gravity separation, or direct manual separation of electrodes and collectors. Leaching and high-temperature treatments remove impurities and binders (Liu et al., 2022).

Recycling graphite used in lithium-ion battery anodes is not yet commercially viable. Graphite degradation occurs chemically, from electrolyte infiltration, or physically, during charge–discharge cycles. Mildly degraded graphite can be re-graphitized for reuse after acid leaching, though at higher costs than natural graphite. Severely degraded graphite is recycled for uses like reducing agents, adsorbents, or high-value graphene production (Liu et al., 2022).

However, as demand for graphite grows and the production of primary resources remains limited, developing recycling technologies to secure secondary resources is becoming essential. Currently, research into recovering graphite from battery anodes is active, and more companies are contributing to advancements in this field including SungEel HiTech, EcoGraf, POSCO, and Ascend Elements.

4.4 Environmental issues in the graphite supply chain

The graphite supply chain faces significant environmental challenges, including carbon emissions, the use of strong acids and bases during purification, and pollution caused by graphite micro-particles during spheroidization. In the natural graphite supply chain, production in China emits an average of 9.6 kg CO₂ equivalent per kg of coated spherical graphite. These emissions primarily come from coating (41%), beneficiation (21%), spheroidization (20%), purification (14%), and mining (4%) (Engels et al., 2022). Outside China, emissions are reported to be lower, ranging from 3 to 7 kg CO₂ equivalent per kg (Fastmarkets, 2024). On the other hand, synthetic graphite production generates far higher emissions due to the high-temperature graphitization process, emitting 20–25 kg CO₂ equivalent per kg of coated spherical graphite (Fastmarkets, 2024).

Purification of natural graphite relies on acid treatments to remove metal oxides and silicate impurities. The process uses strong acids and bases, particularly hydrofluoric acid, to dissolve silicates. This poses serious environmental risks, such as soil and water contamination from wastewater (Jara et al., 2019). Additionally, the graphite mining and processing release fine graphite particles, which has potential pollute both air and water (Golly et al., 2015; Whoriskey, 2016). These environmental impacts make it essential to address sustainability issues in the graphite supply chain.

5 Risks in graphite supply chain

Graphite has become a major topic of concern due to its high supply chain risks (IRENA and NUPI, 2024b). The International Energy Agency (IEA) assessed these risks and classified graphite, along with lithium, as one of the most supply-chain-vulnerable critical minerals among the six

critical minerals: copper, lithium, nickel, graphite, cobalt, and rare earth elements (REE). The rationale for this classification is detailed in Table 3 (IEA, 2024).

According to quantitative supply projections, graphite production can meet future demand, and the market is relatively stable with low price volatility. However, the concentration of supply chains, lack of market transparency, and potential price increases due to environmental issues during production pose significant challenges.

From a supply perspective, graphite is not inherently scarce, meaning its supply risk is relatively low. However, its highly concentrated supply chain makes it a geopolitically vulnerable mineral. As shown in Fig. 7, China dominates over 70% of the graphite supply chain and produces 93% of battery anode materials (or called active anode material, AAM) (Fastmarkets, 2024; SNE Research, 2024).

Graphite's highly concentrated supply chain has placed it at the center of trade restrictions and geopolitical tensions. In December 2023, the Chinese government imposed export controls on graphite, underscoring its strategic importance. While the immediate impact on supply was not severe, the move heightened concerns about the fragility and risks within the global graphite supply chain. In May 2024, the U.S. extended tax credits under the Inflation Reduction Act (IRA) for electric vehicles (EVs) containing Chinese graphite by two years, through the end of 2026, as a measure to address immediate supply chain vulnerabilities. This decision signifies a formal acknowledgment by the U.S. government of its substantial reliance on China for anode-grade graphite, highlighting the current lack of viable alternative supply chains to meet growing domestic demand.

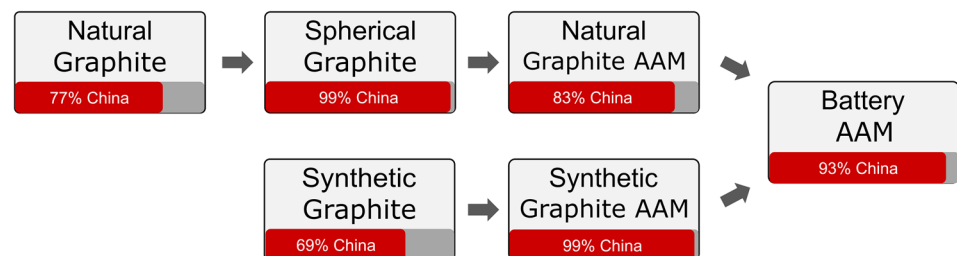
In Europe, the Critical Raw Materials Act (CRMA) seeks to reduce dependency on single-country suppliers for critical materials, including graphite. However, achieving this diversification remains a significant challenge, given China's dominance in graphite mining and processing. Furthermore, escalating trade disputes between the U.S. and China, such as increasing tariffs, continue to exacerbate uncertainties in the graphite supply chain. These geopolitical factors, combined with rising demand for EVs and energy storage systems, underline the urgent need for a more resilient and diversified graphite supply network.

6 How to strengthen the graphite supply chain

Strengthening the graphite supply chain requires a comprehensive approach to address current vulnerabilities and prepare for future challenges (KSME, 2024). Five key recommendations are proposed to strengthen and diversify the supply chain, enhancing its resilience and sustainability within both geological and market contexts.

Table 3 Risk assessment of graphite supply chain (IEA, 2024)

Area	Score	Rationale
Supply risk: low		
Pace of demand growth	High	Projected pace of future demand growth is significantly higher than the rate observed in the 2010s
Short-term market balances	Low	Sufficient supply expected to serve short-term demand
Long-term market balances	Medium	Expected base case mine supply meets two-thirds of the primary supply requirements in 2040, although synthetic graphite could fill the gap
Observed price volatility	Low	Historical price movement has been relatively less volatile
Impact on clean energy cost	Low	Graphite is not a major part of total battery pack costs
Geopolitical risk: high		
Geographical concentration: mining	High	Expected top 3 share in 2030 is around 88% and top 1 share is around 69%
Geographical concentration: refining	High	Battery-grade supply continues to be dominated by China, with 93% share (down from 100% today)
N-1 supply and demand balances	High	Very limited supply potential to serve N-1 battery-grade graphite requirements in 2030
Export risks of major suppliers	High	Risk score 4.4 (mining 4.5/refining 4.4) (OECD, 2024)
Hurdles for diversification	High	New projects are capital-intensive, requiring up to two times higher capital requirements than Chinese projects. New graphite project developers have limited financial capacity
Barrier to respond to disruption: high		
Visible stock levels	Medium	Limited information about visible stock levels, but accumulated industry stocks are estimated to exist
Transparency of pricing schemes	High	Price reporting driven through survey pricing and unregulated platforms with limited liquidity
Availability of options to moderate demand	Medium	Silicon could take growing shares in anodes, but unlikely to challenge graphite use in the near term
Status of secondary supply	High	Share of secondary supply remains low, with limited ability to scale up secondary supply quickly
Exposure to ESG & climate risk: high		
Environmental performance: mining	Medium	Weighted average score 41 (Block et al., 2024)
Environmental performance: refining	High	577 g CO ₂ /kWh
Social and governance performance	High	Weighted average score 0.3 (Coppedge et al., 2024)
Exposure to water stress	Medium	Many natural graphite mines in Heilongjiang and Inner Mongolia, regions with medium water stress
Exposure to earthquake risks	Low	Heilongjiang and Inner Mongolia are not exposed to major earthquake risks

Fig. 7 China's Production Share by Stage in the Graphite Supply Chain (Fastmarkets, 2024; SNE Research, 2024)

6.1 Expanded discovery and mapping of potential graphite deposits

A robust graphite supply chain begins with a reliable and diversified resource base. Identifying and mapping additional graphite deposits with commercial potential is critical. By systematically exploring under-explored areas with

favorable geological conditions for graphite formation, domestic production capacity can be strengthened, reducing reliance on imports. Advanced geological surveys and geochemical analyses can precisely identify regions with high graphite potential, enabling the development of a reliable inventory of graphite resources. Comprehensive

mapping of these deposits will aid in better resource management and strategic planning.

6.2 Advanced graphite exploration technologies

Investing in advanced, multi-faceted exploration technologies is essential to optimize exploration activities. Techniques such as electromagnetic (EM) surveys, induced polarization (IP), and resistivity surveys, combined with remote sensing and advanced geochemical analysis, can improve both accuracy and cost-efficiency. Incorporating machine learning algorithms into data analysis can help identify subtle geological patterns indicative of graphite deposits. Using these technologies in combination can precisely locate high-quality graphite deposits, reducing exploration time, cutting costs, and increasing discovery success rates.

6.3 Cost-effective mining technologies for small deposits

Due to the geological characteristics of graphite, deposits often occur in irregular shapes or steep inclines, following surrounding folded terrains. Many are found as small lenticular bodies, making them challenging and inefficient to mine with traditional large-scale methods. Developing low-cost and accessible mining technologies tailored to small-scale deposits can improve efficiency and make previously uneconomical deposits viable for development. Such technologies should align with the scale and budget of smaller mines, enabling more widespread and efficient graphite resource development.

6.4 Enhanced government support for new graphite projects

Despite the growing demand for graphite, projects in this sector often encounter substantial economic and operational challenges, including high initial investment costs, regulatory hurdles, and market volatility. Government support plays a pivotal role in addressing these challenges, particularly for new entrants to the graphite industry. Policies such as subsidies, tax incentives, and grants can enhance the financial viability of graphite projects, attract private investment, and shorten project development timelines.

Moreover, government-funded infrastructure developments—such as improved transportation networks and reliable power supply in proximity to graphite mines—can significantly reduce operating costs and improve supply chain efficiency. Such initiatives not only support the growth of

the graphite industry but also strengthen its contribution to critical sectors like battery production and the broader energy transition.

6.5 Standardization of HS codes for graphite material tracking

The graphite trade currently faces challenges arising from inconsistent Harmonized System (HS) codes. The existing HS code for graphite is outdated and limited to distinguish graphite and its precursors intended for battery applications. Standardizing HS codes to categorize different types and grades of graphite such as synthetic graphite, flake graphite, and battery anode material could streamline trade procedures, enhance supply chain transparency, and improve the tracking of graphite imports and exports. Effective collaboration with international trade organizations and coordination with industry stakeholders will be essential for successful implementation.

7 Conclusion

Graphite is indispensable for the clean energy transition, serving as the main anode material in lithium-ion batteries. However, its supply chain faces significant risks due to high concentration of production, low market value, and geopolitical uncertainties. A diversified strategy including advanced exploration, cost-effective mining technologies, and standardized trade codes can enhance both sustainability and resilience in the graphite sector. By pursuing these initiatives, stakeholders from governments and research institutions to private investors and industry partners can tackle existing challenges and reinforce the graphite supply chain. Such collaborative efforts will bolster graphite production capacity, strengthen supply security, and ultimately contribute the global shift toward cleaner, more sustainable energy solutions.

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Data availability The data supporting the findings of this study are available upon request. Please contact Dr. Seong-Jun Cho at mac@kigam.re.kr for access to the data.

Declarations

Conflict of interest The authors declare no conflict of interest.

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